



BAULKHAM HILLS HIGH SCHOOL

2011

**TRIAL HIGHER SCHOOL CERTIFICATE
EXAMINATION**

Mathematics Advanced

General Instructions

- Reading time – 5 minutes
- Working time – 3 hours
- Write using black or blue pen
- Board-approved calculators may be used
- A table of standard integrals is provided at the back of the cover sheet.
- All necessary working should be shown in every question
- Marks may be deducted for careless or badly arranged work

Total marks – 120

- Attempt Questions 1 – 10
- All questions are of equal value
- Start each question on a new sheet of paper.
- Write your student number and the question number at the top of each sheet.

STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln(x + \sqrt{x^2 - a^2}), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln(x + \sqrt{x^2 + a^2})$$

NOTE: $\ln x = \log_e x$, $x > 0$

Question 1 (12 marks)

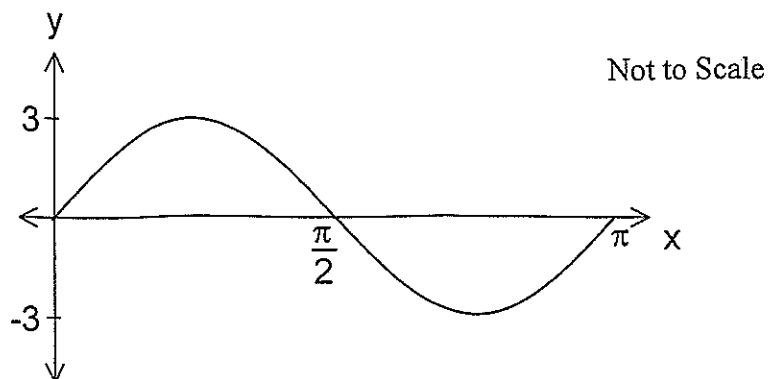
Marks

a) Solve $|3x - 2| < 1$

2

b) The graph below can be expressed in the form $y = A \sin(nx)$

2



What are the values of A and n ?

c) Find the values of a and b such that $\frac{2}{\sqrt{10}+3} = a + b\sqrt{10}$

2

d) Does the series below have a limiting sum? Justify your answer.

2

$$2 - 3 + \frac{9}{2} - \frac{27}{4} \dots \dots \dots$$

e) Differentiate $\frac{e^{2x}}{x}$

2

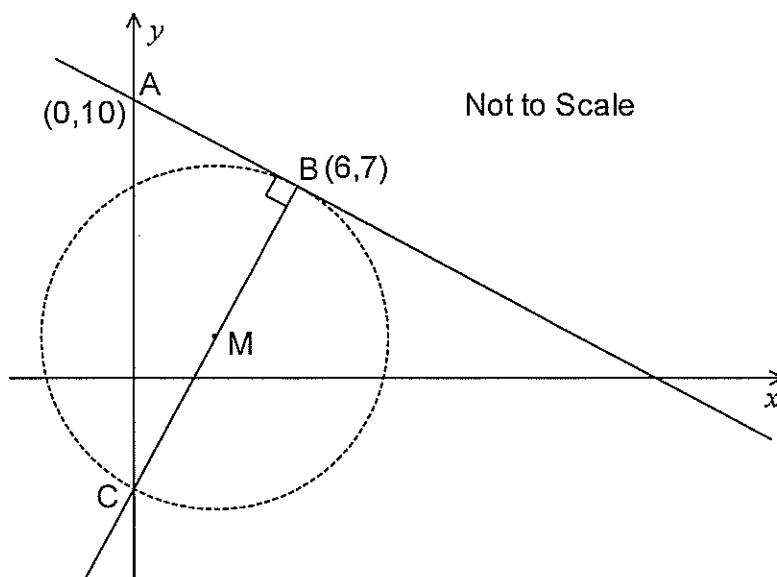
f) Without sketching when is the curve $y = x^3 - 6x^2 + 9x + 2$ increasing?

2

Question 2 (12 marks) - Start on a new page

Marks

a)



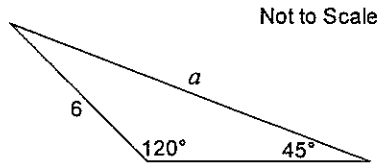
- (i) Find the gradient of AB. 1
- (ii) Show the equation of the line l perpendicular to AB passing through B is given by $y = 2x - 5$. 2
- (iii) The line l cuts the y axis at C. Find the co-ordinates of C. 1
- (iv) M is the midpoint of BC. Find the co-ordinates of M. 1
- (v) Find the equation of the circle which passes through B and C with centre M. 2
- (vi) Another line $y = mx + b$ perpendicular to AB is a tangent to the circle in (v). If $b > 0$ find the point of intersection of this tangent to the circle. 1

- b) (i) Fred weighed 130 kg and lost 1.2 kg every week for a period of time. How much did Fred weigh after n weeks? 1
- (ii) Dana started losing weight at the same time as Fred. She weighed 120.1 kg and lost 0.9 kg each week for a period of time. After how many weeks did Fred and Dana weigh the same? 2
- (iii) What percentage of Fred's body weight had he then lost? 1

Question 3 (12 marks) - Start a new page

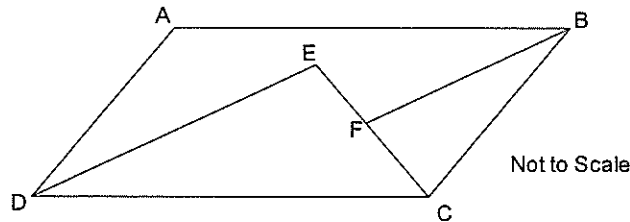
Marks

- a) 2



Find the exact value of a in the diagram above.

- b) 3



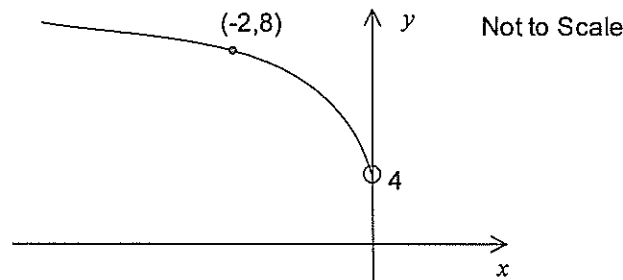
ABCD is a parallelogram. DE bisects $\angle ADC$, EC bisects $\angle DCB$ and BF bisects $\angle ABC$

- (i) Prove $\triangle DEC \parallel \triangle BFC$ 3
 (ii) Show that $\angle DEC = 90^\circ$ 2
 (iii) If $DE = 6\text{ cm}$, $EC = 10\text{ cm}$ and $EF = FC$, find the length of BC. 2
- c) If $\log_a 3 = x$ and $\log_a 2 = y$ express $\log_a 18$ in terms of x and y . 1
- d) If α and β are the roots of $3x^2 - 2x + 6 = 0$ find without solving : 2

$$\frac{1}{\alpha^2\beta} + \frac{1}{\beta^2\alpha}$$

Question 4 (12 marks) - Start a new page

- a) 1



Above is the portion of the curve $y = f(x)$.

- (i) If the function is odd sketch the entire graph of $y = f(x)$ on your own paper. 1
 (ii) State the range of the function. 1
 (iii) If $f(-2) + f(-3) = 17$ find $f(3)$. 2
- b) (i) Find $\int \sqrt{6x+1} dx$ 2
 (ii) Evaluate $\int_0^1 4e^{2x} dx$ 3
- c) Find the equation of the tangent to the curve $y = 2 \cos \pi x$ at $x = \frac{1}{6}$ 3

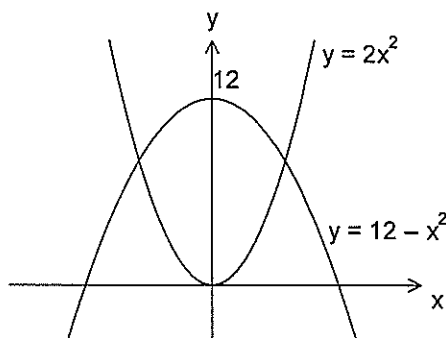
Question 5 (12 marks) - Start a new page

Marks

a) Consider the parabola $x^2 = 4(y - 5)$.

- | | |
|---|---|
| (i) Write down the co-ordinates of the vertex. | 1 |
| (ii) What are the co-ordinates of the focus? | 1 |
| (iii) Sketch the parabola $x^2 = 4(y - 5)$ | 1 |
| (iv) Calculate the area enclosed by the parabola and the line $y = 6$. | 3 |

b)



- | | |
|--|---|
| (i) Show that the points of intersection of the two curves are $(-2, 8)$ and $(2, 8)$. | 1 |
| (ii) The area enclosed by the two curves is rotated about the y axis.
Find the volume of the solid generated. | 5 |

Question 6 (12 marks) - Start a new page

- | | |
|---|---|
| a) (i) Two dice are rolled and the lowest number of the two dice is recorded.
What is the probability that a 2 is recorded when the two dice are rolled? | 2 |
| (ii) Two dice are rolled again and the numbers on both dice differ by 3.
What is the probability that a 2 is recorded? | 2 |

b) (i) Complete the table below for $y = \sqrt{\sin x}$

x	0	$\frac{\pi}{8}$	$\frac{\pi}{4}$	$\frac{3\pi}{8}$	$\frac{\pi}{2}$
y			0.841		

- | | |
|---|---|
| (ii) Hence estimate $\int_0^{\frac{\pi}{2}} \sqrt{\sin x} dx$ using the trapezoidal rule with 4 strips. | 3 |
|---|---|

c) (i) Show that $\frac{d}{dx}(\operatorname{cosec} x) = -\cot x \operatorname{cosec} x$	2
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- | | |
|---|---|
| (ii) Hence evaluate $\int_{\frac{\pi}{6}}^{\frac{\pi}{4}} \cot x \operatorname{cosec} x dx$ | 2 |
|---|---|

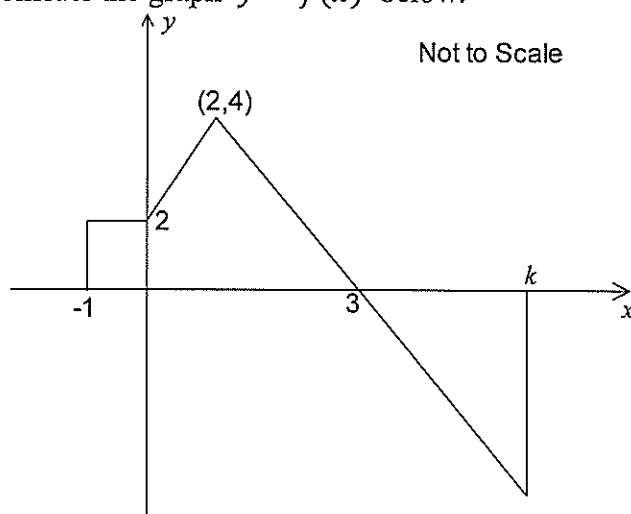
Question 7 (12 marks) - Start a new page

Marks

- a) A farm harvested 3000 tonnes of wheat in the year 2000.
Each year the amount harvested is 4% more than the previous year.
- (i) How much will be harvested in 2020? 2
- (ii) How much will be harvested from 2000 to 2020? 2
- b) A particle moving along the x axis starts at the origin. At time t seconds the particle has a displacement x metres from the origin and is travelling with a velocity of $v \text{ ms}^{-1}$.
The displacement is given by
$$x = 4t - 6\ln(t + 1).$$
- (i) Find an expression for v the velocity of the particle. 1
- (ii) Find the initial velocity of the particle. 1
- (iii) Find when the particle comes to rest. 1
- (iv) Find the distance travelled by the particle in the first 3 seconds. 3
- (v) When is the acceleration of the particle positive? 2

Question 8 (12 marks) - Start a new page

- a) (i) Show that $\frac{d}{dx}[(3-x)(x-1)^3] = 2(x-1)^2(5-2x)$ 2
- (ii) Given $y = (3-x)(x-1)^3$ state the intercepts. 1
- (iii) Find the stationary points and determine their nature. 3
- (iv) Sketch the curve $y = (3-x)(x-1)^3$. 2
- b) Consider the graph $y = f(x)$ below. 4



If $\int_{-1}^k f(x) dx = 0$ find the value of k .

Question 9 (12 marks) - Start a new page**Marks**

- a) A drug is used to control a medical condition. It is known that the quantity Q of the drug remaining in the body after t hours satisfies the equation

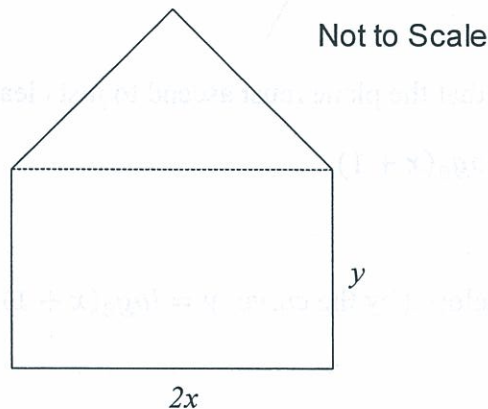
$$Q = Q_0 e^{-kt} \quad \text{where } Q_0 \text{ and } k \text{ are constants.}$$

- (i) An initial dose is administered and 4 hours later half the original dose remains. 2

Find the value of k .

- (ii) What percentage of the initial dose remains after 6 hours? 2

- b) A window is made up of a rectangle surmounted by an equilateral triangle with dimensions as shown. The perimeter of the window is 18 metres.



- (i) Show that the area (A) of the window is given by 2

$$A = 18x - x^2(6 - \sqrt{3})$$

- (ii) Hence find the dimensions of the window that would allow the maximum amount of light to enter the window. Give your answer to the nearest centimetre. 4

- c) If $f(x) = x + \frac{1}{x}$ show that $f(x) \cdot f\left(x + \frac{1}{x}\right) = f(x^2) + 3$ 2

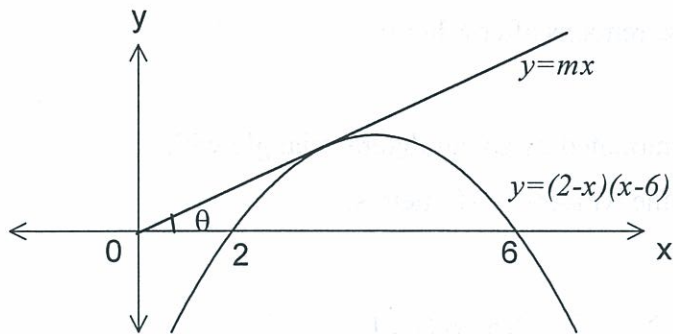
Question 10 (12 marks) - Start a new page

a) Solve $4 \tan x \sin x + 3 \sin x - \cos x = 0$ for $0^\circ \leq x \leq 360^\circ$

3

- b) The line $y = mx$ represents the flight path of a plane which has just taken off from the airport at 0. The parabola $y = (2 - x)(x - 6)$ represents a hill that a plane must fly over.

4



Find the angle of elevation θ that the plane must ascend to just clear the hill.

- c) (i) Sketch the graph $y = \log_8(x + 1)$

1

- (ii) Differentiate $x = 8^y$

1

- (iii) Hence find the area enclosed by the curve $y = \log_8(x + 1)$, the x axis and the line $x = 7$.

3

End of Exam

1a) $|3x-2| < 1$
 $3x-2 < 1 \quad -3x+2 < 1$
 $x < 1 \quad -3x < -1$
 $x > \frac{1}{3}$ ①

Soln: $\frac{1}{3} < x < 1$ ①
 Aw ① for either $x < 1$ or $x > \frac{1}{3}$

b) $A=3 \quad \frac{2\pi}{n} = \pi \Rightarrow n=2$ ①

c) $\frac{2}{(\sqrt{10}+3)(\sqrt{10}-3)} = \frac{2\sqrt{10}-6}{10-9}$
 $= -6+2\sqrt{10}$
 $\Rightarrow a=-6 \quad b=2$ ①

Aw ① for rationalising correctly

Aw ① for error but a & b correct from previous working (CPW)

d) $r = \frac{T_2}{T_1} = -\frac{3}{2}$ ①

Since $-\frac{3}{2} \neq 1$ then ①
 the series does not have a limiting sum (Must conclude correctly)

2011 final 2U solutions

e) $\frac{d}{dx} \left(\frac{e^{2x}}{x} \right) = \frac{2xe^{2x} - e^{2x}}{x^2}$ ②
 $= \frac{e^{2x}(2x-1)}{x^2}$ Not necessary

Aw ① for $\frac{d}{dx}(e^{2x}) = 2e^{2x}$

Aw ① for correct use of formula with 1 error.

f) y increases when $y' > 0$

ie $3x^2 - 12x + 9 > 0$
 ie $3(x^2 - 4x + 3) > 0$ ①

$3(x-1)(x-3) > 0$

$\therefore x < 1$ or $x > 3$ ①

2. (i) $m_{AB} = \frac{10-7}{0-6} = -\frac{1}{2}$ ①

(ii) $\perp m = 2$ ①

$y-7 = 2(x-6)$

$y = 2x - 12 + 7$

$y = 2x - 5$ ①

(iii) at C $x=0$ ①

$\therefore y = -5 \quad C(0, -5)$

Aw ① for just $y = -5$.

(iv) $M\left(\frac{0+6}{2}, \frac{-5+7}{2}\right) = (3, 1)$ ①

(v) $MB = \sqrt{(6-3)^2 + (7-1)^2}$
 $= \sqrt{45}$ ①

eqn $(x-3)^2 + (y-1)^2 = 45$ ①

(vi) To find the point from M move $\leftarrow$ 6 units $\uparrow$ 3. point is $(-3, 4)$ ①

b) Fred = $130 - 1.2n$ ①

(i) $130 - 1.2n = 120 - 0.9n$
 $9.9 = 0.3n$ ①

$n = 33$ ①

(ii) Fred weighed $130 - 1.2 \times 33$
 $= 90.4 \text{ kg}$

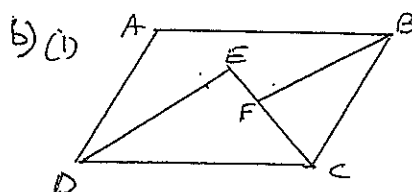
% loss = $\frac{39.6}{130} \times 100\%$
 $= 30.46\%$ ①

3. a) $\frac{a}{\sin 120^\circ} = \frac{6}{\sin 45^\circ}$

$a = \frac{6^3 \cdot \frac{\sqrt{3}}{2}}{\frac{1}{\sqrt{2}}}$
 $= 3\sqrt{6}$

Aw ① for correct sub'n into sine rule.

Aw ① for correct exact values & wrong final answer



Let $\angle ADE = x^\circ$

$\therefore \angle EDC = x^\circ$ (DE bisects $\angle ADC$)

$\therefore \angle ABC = 2x^\circ$ (opposite $\angle$'s of par'm)

$\therefore \angle FBC = x^\circ$ (BF bisects $\angle ABC$)

Let $\angle ECD = y^\circ$

$\therefore \angle ECB = y^\circ$ (EC bisects $\angle BCD$)

$\therefore \triangle DEC \parallel \triangle BFC$ (AA)

Aw ① for using fact that diagonals bisect angles.

Aw ① for opposite $\angle$'s of par'm =

Aw ① for correct similarity test.

(ii) $\angle ADC + \angle BCD = 180^\circ$ (Co-interior $\angle$'s on $\parallel$ lines)

$\therefore 2x + 2y = 180^\circ$

$\therefore x + y = 90^\circ$

hence $\angle DEC = 90^\circ$ (Angle Sum of Δ) ①

(iii) $\frac{EC}{FC} = \frac{DE}{BF}$ (sides in ratio in $\parallel \Delta$'s)

ie $\frac{10}{5} = \frac{6}{BF}$ } ①

$\therefore BF = 3$

$\therefore BC^2 = BF^2 + FC^2$

$BC = \sqrt{3^2 + 5^2}$
 $= \sqrt{34}$ } ①

c) $\log_a 3 = x \quad \log_a^2 = y$

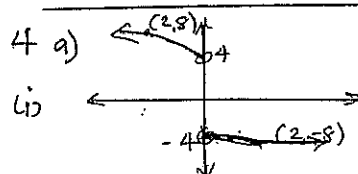
$\therefore \log_a 18 = 2 \log_a 3 + \log_a^2$

d) $3x^2 - 2x + 6 = 0$

$\alpha + \beta = \frac{2}{3} \quad \alpha\beta = 2$ ①

$\therefore \frac{1}{\alpha^2\beta} + \frac{1}{\beta^2\alpha} = \frac{\beta + \alpha}{\alpha^2\beta^2}$
 $= \frac{\frac{2}{3}}{2^2}$
 $= \frac{1}{6}$ ①

Aw ① if error in $\alpha + \beta$ or $\alpha\beta$ but CPW



(ii) Range $y > 4$ or $y < -4$

(iii) $f(-2) + f(-3) = 17$

then $f(2) + f(3) = -17$ ①

$\therefore -8 + f(3) = -17$
 $\therefore f(3) = -9$ ①

b(i) $\int (6x+1)^{\frac{1}{2}} dx$
 $= \frac{(6x+1)^{\frac{3}{2}}}{\frac{3}{2}} + C$

① $\leftarrow \frac{3}{2} \times 6$
 $= \frac{1}{9} (6x+1)^{\frac{3}{2}} + C$ (ignore c)

$$\begin{aligned} \text{biv } \int_0^1 4e^{2x} dx &= 2[e^{2x}]_0^1 \quad (1) \\ &= 2(e^2 - e^0) \quad (1) \\ &= 2(e^2 - 1) \quad (1) \end{aligned}$$

Aw(1) for incorrect $\int$ but CPW

$$c) y = 2\cos\pi x$$

$$\text{at } x = \frac{1}{6} \quad y = 2\cos\frac{\pi}{6} \Rightarrow y = \sqrt{3} \quad (1)$$

$$y' = -2\pi \sin\pi x$$

$$\text{at } x = \frac{1}{6} \quad y' = -2\pi \sin\frac{\pi}{6} = -\pi \quad (1)$$

$$\therefore \text{Eqn } y - \sqrt{3} = -\pi(x - \frac{1}{6}) \quad (1)$$

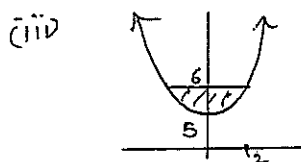
$$y = -\pi x + \frac{\pi}{6} + \sqrt{3} \quad (1)$$

$$5 a) x^2 = 4(y-5)$$

$$(i) \text{ Vertex } (0, 5) \quad (1)$$

$$(ii) 4a = 4 \quad a = 1$$

$$\therefore S(0, 6) \quad (1)$$



$$(iii) \text{ when } y = 6 \quad x = \pm 2$$

$$x^2 = 4(y-5) \Rightarrow y = \frac{x^2}{4} + 5$$

$$\therefore \text{Area} = \int_{-2}^2 6 - (\frac{x^2}{4} + 5) dx$$

$$= 2 \int_0^2 1 - \frac{x^2}{4} dx \quad (1)$$

$$= 2 \left[x - \frac{x^3}{12} \right]_0^2 \quad (1)$$

$$= 2 \left[2 - \frac{8}{12} - 0 \right]$$

$$= \frac{8}{3} \quad (1)$$

$$b) 12 - x^2 = 2x^2$$

$$\therefore 3x^2 = 12$$

$$x = \pm 2$$

$$y = 2(\pm 2)^2 \rightarrow y = 8$$

$$\therefore \text{pts } (2, 8) \quad (-2, 8)$$

$$(ii) y = 2x^2 \rightarrow x^2 = \frac{y}{2}$$

$$y = 12 - x^2 \rightarrow x^2 = 12 - y$$

$$\therefore \text{Volume} =$$

$$\pi \int_0^8 \frac{y}{2} dy + \pi \int_8^{12} 12 - y dy$$

$$(1) \text{ for correct bounds}$$

$$(1) \text{ for correct substitution for } x$$

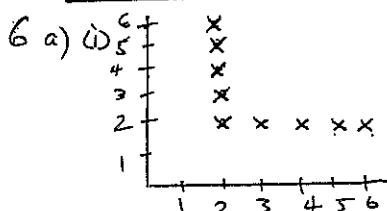
$$(1) \text{ for summing volumes}$$

$$= \pi \left[\frac{y^2}{4} \right]_0^8 + \pi \left[12y - \frac{y^2}{2} \right]_8^{12} \quad (1)$$

$$= \pi [16 - 0] + \pi [(44 - 72) - (96 - 32)] \quad (1)$$

$$= 16\pi + 8\pi$$

$$= 24\pi \quad (1)$$



x denotes 2 as lowest score

$$\therefore P(2 \text{ recorded}) = \frac{9}{36} \rightarrow (1)$$

$$(1) \leftarrow \frac{1}{4}$$

6a) (i) Sample Space

$$\{(1, 4), (2, 5), (3, 6), (4, 1), (5, 2), (6, 3)\} \quad (1)$$

$$\therefore P(\text{recording 2}) = \frac{2}{6} = \frac{1}{3} \quad (1)$$

x	0	$\frac{\pi}{8}$	$\frac{\pi}{4}$	$\frac{3\pi}{8}$	$\frac{\pi}{2}$
y	0	0.619	0.841	0.961	1

$$y = \sqrt{\sin x} \quad (1)$$

$$(ii) \text{ Area} = \frac{\pi}{8} \left[0 + 1 + 2(0.619 + 0.841 + 0.961) \right] \quad (1)$$

$$= 1.147 \rightarrow (1)$$

$$c (i) \frac{d}{dx} (\sin x)^{-1} = -(\sin x)^{-2} \cos x \quad (1)$$

$$(1) \left\{ \begin{aligned} &= \frac{-\cos x}{\sin^2 x} \\ &= -\cot x \operatorname{cosec} x \end{aligned} \right.$$

$$(ii) \int_{\frac{\pi}{6}}^{\frac{\pi}{4}} \cot x \operatorname{cosec} x$$

$$= -(\operatorname{cosec} x) \Big|_{\frac{\pi}{6}}^{\frac{\pi}{4}} \quad (1)$$

$$= -\left[\operatorname{cosec} \frac{\pi}{4} - \operatorname{cosec} \frac{\pi}{6} \right]$$

$$= -(\sqrt{2} - 2)$$

$$= 2 - \sqrt{2} \quad (0.585) \quad (1)$$

$$7 a) (i) 3000 \times 1.04^{20} = 6573.46 \quad (1)$$

$$(ii) \text{ Total} = 3000 + 3000 \times 1.04^1 + \dots + 3000 \times 1.04^{20} \quad (1)$$

$$S_n = \frac{3000(1.04^{21} - 1)}{1.04 - 1} \quad (1)$$

$$= 95907.6 \quad (1)$$

$$b) x = 4t + 6 \ln(t+1)$$

$$(i) v = 4 - \frac{6}{t+1} \quad (1)$$

$$(ii) \text{ when } t=0 \quad v = -2 \text{ m/s} \quad (1)$$

$$(iii) v=0 \quad 4 = \frac{6}{t+1}$$

$$t+1 = \frac{3}{2}$$

$$t = \frac{1}{2} \quad (1)$$

$$(iv) \text{ when } t=0 \quad x=0$$

$$(1) t = \frac{1}{2} \quad x = 2 - 6 \ln \frac{3}{2}$$

$$= (-0.433)$$

$$\text{when } t=3 \quad x = 12 - 6 \ln 4 \quad (1)$$

$$= (3.682)$$

$$\therefore \text{Distance} = 2x| -0.433| + 3(6.82)$$

$$= 4.548 \quad (1)$$

$$(v) a = \frac{d}{dt} (4 - 6(t+1)^{-1})$$

$$= \frac{6}{(t+1)^2} \quad (1)$$

$$\left\{ \begin{aligned} &\text{since } (t+1)^2 > 0 \text{ for all } t \\ &\text{then } \frac{6}{(t+1)^2} > 0 \text{ for all } t \end{aligned} \right.$$

$$\therefore \text{accel'n is always positive} \quad (1)$$

$$8. \frac{d}{dx} ((3-x)(x-1)^3)$$

$$(i) \text{ abn } = -1(x-1)^3 + 3(x-1)^2(3-x)$$

$$= (x-1)^2 [-(x-1) + 3(3-x)]$$

$$= (x-1)^2 [-x+1+9-3x]$$

$$= (x-1)^2 [-2x+10]$$

$$= (x-1)^2 (10-2x) \quad (1)$$

$$= \frac{2(x-1)^2 (5-x)}{2}$$

$$(ii) \text{ at } y=0 \quad (3-x)(x-1)^3 = 0$$

$$\therefore x = 3, 1 \quad (1)$$

$$\text{when } x=0 \quad y = -3$$

$$(ii) \text{ st pts } 2(x-1)^2(5-2x) = 0$$

$$(1) \left\{ \begin{aligned} &x = 1 \\ &y = 0 \end{aligned} \right. \quad \frac{5}{2}$$

Test.
Nature using y'

$x=1$

x	0	1	2
y'	10 ↗	0 ↔	2 ↘

①

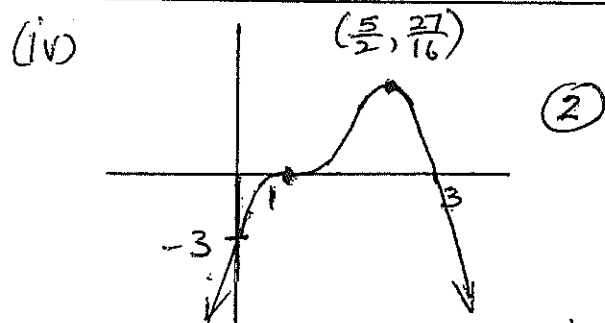
∴ Horizontal pt. of inflex.

$x=2\frac{1}{2}$

x	2	$2\frac{1}{2}$	3
y'	2 ↗	0 ↔	-2 ↘

①

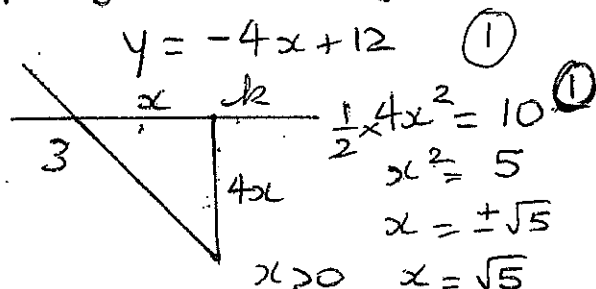
∴ Max t.p.



Must show intercepts & st. pts.

b) Area above x axis = 10 unit²
①

eq'n of line through (2,4)(3,0)



∴ $k = 3 + \sqrt{5}$ ①

a) ① $Q = Q_0 e^{-kt}$
when $t=4 \quad Q = \frac{Q_0}{2}$

∴ $\frac{Q_0}{2} = Q_0 e^{-4k}$ ①

$\frac{1}{2} = e^{-4k}$

∴ $\ln \frac{1}{2} = -4k \cdot \ln e$

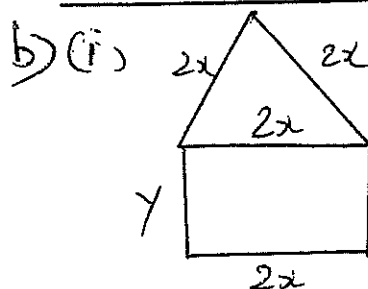
∴ $k = \frac{\ln \frac{1}{2}}{-4}$

$= 0.173 \dots$ ①

(ii) when $t=6$
 $Q = Q_0 e^{-6(0.173 \dots)}$ ①

$= Q_0 \times 0.3535 \dots$

∴ 35.4% of the original dose remains ①



$A = 2xy + \frac{1}{2} \cdot (2x)^2 \sin 60$
 $= 2xy + 2x^2 \cdot \frac{\sqrt{3}}{2}$ ①

$A = 2xy + x^2\sqrt{3}$

but $6x + 2y = 18$

∴ $y = 9 - 3x$

∴ $A = 2x(9 - 3x) + x^2\sqrt{3}$
 $= 18x - 6x^2 + x^2\sqrt{3}$ ①
 $= 18x - x^2(6 - \sqrt{3})$

c) $\frac{dA}{dx} = 18 - 2(6 - \sqrt{3})x = 0$ ①

when $x = \frac{18}{2(6 - \sqrt{3})}$

$= \frac{9}{6 - \sqrt{3}}$ ①

test $A'' = -2(6 - \sqrt{3}) < 0$
 ∴ max t.p. ①

$x = 211 \text{ cm.}$

∴ $2x = 422 \text{ cm}$

$y = 900 - 3(211)$
 $= 267 \text{ cm.}$ ①

c) $f(x) = x + \frac{1}{x}$

$f(x + \frac{1}{x}) = x + \frac{1}{x} + \frac{1}{(x + \frac{1}{x})}$

$f(x) \cdot f(x + \frac{1}{x}) = (x + \frac{1}{x}) \left[(x + \frac{1}{x}) + \frac{1}{(x + \frac{1}{x})} \right]$ ①

$$\begin{aligned}
 &= \left(x + \frac{1}{x}\right)^2 + 1 \\
 &= x^2 + \frac{1}{x^2} + 2 + 1 \\
 &= x^2 + \frac{1}{x^2} + 3 \\
 &= f(x^2) + 3
 \end{aligned}
 \quad \textcircled{1}$$

10a) $4 \tan x \sin x + 3 \sin x - \cos x = 0$

$$4 \frac{\sin^2 x}{\cos x} + 3 \sin x - \cos x = 0$$

$$4 \sin^2 x + 3 \sin x \cos x - \cos^2 x = 0$$

$$(4 \sin x - \cos x)(\sin x + \cos x) = 0$$

$$\therefore 4 \sin x = \cos x \quad \sin x = -\cos x$$

$$\therefore \tan x = \frac{1}{4} \quad \tan x = -1$$

$$x = 14^\circ 2', 194^\circ 2', 135^\circ, 315^\circ$$

b) Need pt of intersection

ie $mx = (2-x)(x-6)$

$$mx = -x^2 + 8x - 12$$

$$\therefore 0 = -x^2 + x(8-m) - 12$$

Since $y = mx$ is a tangent then only 1 point of intersection

$$\therefore \Delta = 0$$

$$\text{ie } (8-m)^2 - 4x - 1x - 12 = 0$$

$$64 - 16m + m^2 - 48 = 0$$

$$m^2 - 16m + 16 = 0$$

$$m = \frac{16 \pm \sqrt{16^2 - 4 \cdot 16}}{2}$$

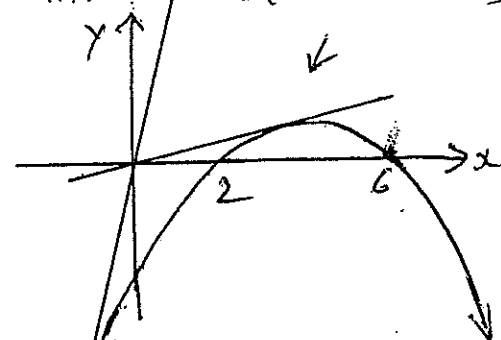
$$\textcircled{1} \quad m = \frac{16 \pm \sqrt{192}}{2}$$

$$m = \frac{16 + \sqrt{192}}{2}, \quad m = \frac{16 - \sqrt{192}}{2}$$

The required gradient is

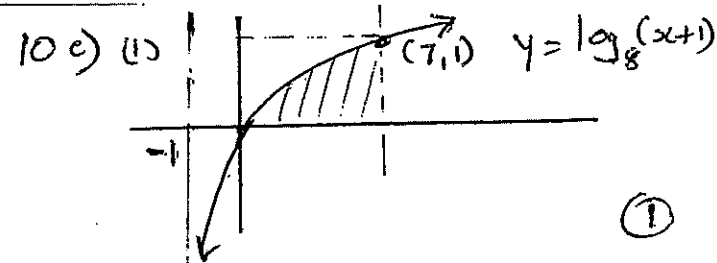
The smallest value of m

Since steeper gradient will intersect below x axis



$$\therefore \tan \theta = \frac{16 - \sqrt{192}}{2}$$

$$\theta = 46^\circ 59'$$



(ii) $x = 8^y$

$$\frac{dx}{dy} = 8^y \ln 8$$

(iii) Area = Rectangle -

$$y = \log_8(x+1)$$

$$x+1 = 8^y$$

$$x = 8^y - 1$$

$$\therefore \text{Area} = 7 \times 1 - \int_0^1 (8^y - 1) dy$$

$$= 7 - \left[\frac{8^y}{\ln 8} - y \right]_0^1$$

$$= 7 - \left[\left(\frac{8}{\ln 8} - 1 \right) - \left(\frac{1}{\ln 8} - 0 \right) \right]$$

$$= 7 - \frac{8}{\ln 8} + 1 + \frac{1}{\ln 8}$$

$$= 8 - \frac{7}{\ln 8} \quad (4.634)$$